The Field-Cell Self: A Falsifiable Mechanistic Identity Hypothesis for Conscious Neural Rendering

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Abstract

The brain contains no anatomically privileged spectator that watches its own activity. Yet an organism maintains a changing point of view, distinguishes body from world, remembers earlier states, considers actions, and encounters their consequences. Neuroscience can identify self-related representations, recurrent processing, body maps, interoceptive signals, and extracellular electrical fields, but that does not by itself identify the physical observer.

This theory-and-methods paper asks whether the continuing observer can be identified with a measurable organization rather than a second entity hidden behind neural signals. The proposed observer is not a cell, anatomical point, abstract record, or electromagnetic field in isolation. It is the continuing organization of a differentiated, recurrent, memory-bearing, embodied process in which living cells generate, receive, preserve, and alter structured bioelectric activity. This falsifiable companion hypothesis within Self-Aware Networks (SAN) is called the **Field-Cell Self**.

The proposal has four claim levels. Level 1 contains independently supported components: neural currents generate extracellular fields; bounded fields can alter neural timing and excitability; neurons perform state-dependent dendritic computation; and recurrent, interoceptive, sensorimotor, memory, and self-model processes contribute to behavior. Level 2 joins conventional signaling and eligible local field effects in a receive-transform-re-express loop that updates an embodied self / world model. **Tonic** denotes a maintained context, while a **phase-wave differential** (PWD) is a preregistered, receiver-relative change in phase, frequency, amplitude, duration, waveform, or spatial propagation. Level 3 predicts that field-cell variables add content-specific information and causal influence beyond rate, power, connectivity, movement, arousal, and flexible recurrent history. Level 4 separates two questions. A candidate conscious episode requires differentiated content, recurrent causal access, and a receiver-dependent organization that carries and changes that content. A diachronic personal self additionally requires memory, embodiment, perspective, action-return closure, and continuity across episodes. Under those declared conditions, the organized process is hypothesized to be the experience and continuing observer rather than a representation delivered to a hidden reader.

A Shannon-bounded analysis distinguishes a predictable tonic context, the surprisal of a candidate PWD, and the PWD's conditional information about an independently measured target.

The paper specifies a coupled model, an individuation rule, comparisons with major alternatives, and seven discriminating experiments. Existing field effects, decoding, report, and synthetic implementation do not establish the identity claim. It loses support if field-cell variables add no held-out content information, receiver-dependent perturbations fail under matched controls, or established alternatives explain the results without a distinct operation. Even jointly favorable tests support an abductive identity inference, not direct observation or deductive proof.

ephaptic coupling; neural oscillations; phase coding; recurrent processing; embodiment; interoception; Self-Aware Networks; phase-wave differential

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Keywords: consciousness; self; neural rendering; bioelectricity; extracellular electric fields;

Scope and evidence boundary

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Correspondence: Micah Blumberg ### Version 1.2 - Draft 9 final preprint Theory-and-methods preprint. No new human or animal experiment is reported. Reserved DOI for the unpublished deposition: doi:10.5281/zenodo.21399661

1. The problem without an inner spectator

When a person sees a red cup and reaches for it, many operations occur in parallel. Photoreceptors respond to light. Visual pathways estimate edges, color, depth, identity, and motion. Body-state systems represent gaze, head position, limb configuration, balance, need, and expected effort. Memory supplies prior encounters with cups and the present goal. Motor systems prepare and execute a movement. Sensory consequences return from the eyes, muscles, skin, and vestibular organs. None of these operations occurs at one point, yet they can participate in one temporarily coherent episode: *I see the cup; I intend to reach; my hand moves; the cup is now in my grasp.*

The easy but misleading picture places a miniature observer at the end of this process. Neural signals would be assembled into an internal display and someone inside the brain would inspect it. That picture cannot explain observation because the inner observer would require another nervous system and another observer. The regress ends only when sensing, modeling, choosing, and updating are operations of the same physical system.

This is not merely a philosophical puzzle. It determines what a neuroscience of consciousness must measure. A decoder can reveal that an experimenter can recover stimulus identity from neural activity. A self-reference task can reveal activity associated with judging one's own traits. Recurrent processing can reveal that earlier activity affects later activity. None of those results alone identifies the physical observer. They may establish representation, access, report, or control while leaving open why and when the organized process is accompanied by experience.

Self-Aware Networks began from a related premise: the brain is a distributed learning process that predicts, acts, receives the consequences of action, and changes what later signals mean [1-3]. Hebbian assemblies and phase sequences provide an earlier route from coactivity to organized recurrent trajectories [4]. Hofstadter's strange loop provides an earlier account of a system whose symbols and levels fold back upon a representation of "I" [5]. Hawkins's memory-prediction framework emphasizes hierarchical prediction and sensorimotor regularity [6], while later cortical reference-frame work proposes location-conditioned object learning in cortical columns [7]. These are important foundations. The present question is narrower and more physical:

> What process continues from moment to moment as the entity that generates, receives, remembers, and changes an embodied neural rendering?

A candidate answer need not posit a new substance. The paper next asks whether the observer is identical to a particular organization of familiar biological processes plus a specific, testable field-cell relation. Section 2 first separates the inherited components from the new conjecture; only then does the manuscript name the joined proposal.

1.1 Contributions and decision logic

The paper makes four contributions:

1. It separates established electrophysiology, the SAN synthesis, the field-as-content prediction, and the identity hypothesis so evidence cannot migrate silently between claim levels.
2. It separates the eligibility conditions for a candidate conscious episode from the additional memory, embodiment, perspective, action-return, and continuity conditions for a diachronic self.
3. It converts the proposal into numbered models and seven experiments with matched baselines, mediation requirements, maximum licensed conclusions, and failure criteria.
4. It compares SAN with major consciousness and bioelectric frameworks using three possible outcomes: reduction, implementation, or a reproducible empirical increment.

The decision rule is deliberately asymmetric. Established components remain valid if the SAN synthesis fails; the synthesis may remain useful if field-as-content fails; and the functional model may survive if mechanistic identity fails. Stronger claims inherit the burden of every lower level and add their own.

1.2 The distinctive joined SAN signature

The proposed SAN contribution is not any one generic component considered alone. Prediction, recurrence, dendritic computation, oscillations, field effects, embodiment, and self-modeling each have earlier literatures. The comparison object is their ordered conjunction: distributed living receivers generate and re-encounter structured electrical state; receiver-relative changes update an active tonic context; neural arrays receive, transform, and re-express partial renderings; memory and action carry the process forward; and the continuing process itself is the proposed observer. Dated SAN sources contain partial ancestors in 2011, 2012, and 2017, a molecular transmission and computational-unit bridge in 2021, the mature NAPOT / PWD field-cell rendering architecture by 2022, and body-map plus sensorimotor-context refinements in 2024 and 2025. Section 10 states this genealogy atom by atom without backdating the mature terminology or treating chronology as biological evidence.

2. The claim ladder

The word *field* is especially vulnerable to equivocation. It can mean a measured extracellular electric potential, a mathematical state field, a population activity map, a phenomenological field of experience, or an unrestricted metaphysical medium. This paper uses the term physically only for a declared electrical quantity in a declared preparation. The strongest consciousness claim is then stated separately as a hypothesis about the organized field-cell process.

2.1 Level 1: established or bounded components

The following propositions have independent support:

1. Neural transmembrane currents contribute to extracellular electric potentials recorded at multiple scales [15,24].
2. Weak endogenous or applied electric fields can alter membrane potential, spike timing, or network activity in bounded preparations [25-27].
3. Field sensitivity depends on cellular class, morphology, receiver state, and stimulation frequency rather than being one uniform broadcast effect [26,27].
4. Phase relative to ongoing activity can carry information beyond spike count in selected preparations, and relative timing can regulate effective communication [16-18].
5. Dendrites are state-dependent nonlinear integrators, not passive wires [19,20].
6. Feedforward and recurrent processing can make experimentally distinguishable contributions [21-23].

7. Body ownership, self-location, interoceptive processing, self-reference, and action monitoring are measurable and partly dissociable [32-37].
8. Effective connectivity and perturbational complexity differ across selected conscious and unconscious conditions [29-31].

These findings do **not** establish SAN, Neural Rendering, field-carried conscious content, or phenomenal identity.

2.2 Level 2: the SAN field-cell synthesis

Level 2 proposes that ordinary synaptic, axonal, chemical, gap-junction, neuromodulatory, sensorimotor, and body-regulatory routes operate together with eligible local electric-field effects in a recurrent loop:

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cells and circuits receive a structured state
-> their learned and current state transforms it
-> their outputs change other cells, the body, and the extracellular electrical state
-> those consequences become part of later receiver input
-> memory and plasticity alter what later states can mean and do
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SAN calls the repeated construction and revision of an action-usable self / world state **Neural Rendering** [1,3]. Rendering does not imply a literal picture or a spectator. It names a state-update operation whose output must be identified by independently measured perceptual, self-related, predictive, or action variables.

2.3 Level 3: the field-as-content hypothesis

Level 3 predicts that a declared field-cell state adds content-specific information and causal influence beyond strong conventional models. A local field effect is not enough. The target result must survive models given the same spikes, rates, power, waveform, connectivity, recurrent history, movement, arousal, and sensory evidence. A perturbation must change a preregistered content dimension through the predicted receiver state while limiting nonspecific disruption.

2.4 Level 4: the mechanistic identity hypothesis

Level 4 proposes that an appropriately organized field-cell process is not merely correlated with experience or read by another system. The process is the physical observer, and its intrinsic condition constitutes the experience.

This is a bridge hypothesis, not a result derived from Levels 1-3. Its scientific value depends on whether it generates discriminating consequences for conscious versus nonconscious organization, qualitative similarity, self continuity, and causal intervention. The joined proposal is called the **Field-Cell Self** only after its problem, inherited components, operation, and claim ceiling have been stated.

2.5 Explicit exclusions

This paper does not claim that:

- every electric or electromagnetic field is conscious;
- a field in isolation, detached from receiver cells, memory, and embodiment, is a self;
- ephaptic coupling replaces synaptic transmission;
- EEG, ECoG, LFP, or MEG directly measures all current sources or proves field causation;
- neural coherence is quantum coherence;
- magnetic mediation follows from extracellular-voltage evidence;

- every phase difference carries information;
- high surprisal is equivalent to semantic, useful, or conscious information;
- a low conditional information rate makes a tonic context biologically unimportant;
- a decoder reveals what the nervous system causally uses;
- reportability is identical to phenomenal experience;
- one frequency band is permanently tonic and another permanently phasic;
- the brain contains a single master map or central canvas;
- a synthetic implementation is sentient merely because it reproduces behavior; or
- chronology is biological evidence.

2.6 Operational naming rule

This paper uses **extracellular electric potential** or **extracellular electric field** for a measured, reconstructed, or manipulated electrophysiological quantity. It uses **field-cell state** for the joint state of those electrical variables and the living receivers whose membrane, dendritic, synaptic, metabolic, and plastic histories constrain their effects. A mathematical state field is named by its specific model and is not presumed to be an extracellular field. **Phenomenal field** refers only to the philosophical target of a structured domain of experience. Evidence for an extracellular electric effect does not establish a magnetic information channel, a single brain-wide field, or phenomenal identity.

3. From cells to a reciprocal electrical state

3.1 The field is generated by living current sources

Neural electrical recordings are consequences of transmembrane and axial currents filtered by geometry, conductivity, distance, reference scheme, and measurement scale [24]. A minimal forward model is

$$V_e(\mathbf{r},t) = \int_\Omega G(\mathbf{r},\mathbf{r}')I_m(\mathbf{r}',t)\,d\mathbf{r}' + \epsilon(\mathbf{r},t), \tag{1}$$

where I_m is a declared transmembrane-current source density, G is a conductivity- and geometry-dependent forward kernel, and ϵ includes measurement noise and unmodeled sources. Equation (1) is an observation model. It does not imply that the measured potential is itself the causal variable or that the inverse source problem is uniquely solved.

Absolute extracellular potential depends on the reference scheme. The reference-invariant candidate input to a cell is therefore expressed through the electric field

$$\mathbf{E}_e(\mathbf{r},t) = -\nabla V_e(\mathbf{r},t) \tag{1}$$

and through declared morphology-relative differences such as $\Delta V_{e,i}^{ab}(t) = V_e(\mathbf{r}_{i,a},t) - V_e(\mathbf{r}_{i,b},t)$ across two locations on receiver i. These quantities remain model-dependent because conductivity, source geometry, and spatial resolution remain uncertain, but they do not assign biological meaning to an arbitrary recording reference.

The important first fact is reciprocal possibility. Frohlich and McCormick used weak fields and closed-loop feedback in an in-vitro neocortical slow-oscillation preparation and found that fields within a physiological range could enhance or entrain network activity [25]. Anastassiou and colleagues measured subthreshold field effects on cortical pyramidal neurons in slices and reported spike entrainment, particularly for slow fields [26]. Lee and colleagues later found strong cell-class and frequency dependence across major cortical neuronal classes in rodent and human tissue [27]. These results establish neither consciousness nor brain-wide field broadcasting. They establish that, under bounded conditions, neural activity can change an electrical environment that can in turn change neural activity.

Recent causal stimulation further strengthens the experimental opportunity without validating the identity thesis. Traveling-wave transcranial alternating-current stimulation has been used to generate directionally organized electric fields, with intracranial validation, directional modulation of monkey spiking, and direction-dependent behavioral effects in humans [28]. This shows that spatial-temporal field organization can be manipulated and can matter for cognition. It does not show that the manipulated field carried conscious content or constituted an observer.

3.2 Cells are receivers with histories

The effect of an extracellular field is not determined by the field alone. Morphology, orientation, membrane conductance, channel expression, dendritic state, synaptic input, recent firing, and cell class affect the receiver's response [19,20,26,27]. This matters conceptually. A global waveform is not a message with one meaning independent of its receivers. The same perturbation can be weak, effective, phase advancing, phase delaying, or behaviorally irrelevant for different cells and states.

Let $x_i(t)$ denote the fast state of receiver i, $m_i(t)$ its slower learned and plastic state, $s_i(t)$ its conventional synaptic and axonal input, $c_i(t)$ chemical, gap-junction, and neuromodulatory terms, and $b(t)$ measured body-state input. A deliberately general state equation is

$$\dot{x}_i(t) = F_i(x_i(t), m_i(t), s_i(t), c_i(t), b(t)) + \alpha_i H_i\Big[\mathbf{E}_e(\mathcal{N}_i, t), \{\Delta V_{e,i}^{ab}(t)\}_{(a,b)\in\mathcal{G}_i}; \mathbf{o}_i, \rho_i(t)\Big] + \xi_i(t). \quad (2)$$

H_i is an eligible local field-to-cell term over declared neighborhood $\mathcal{N}_i$; $\mathcal{G}_i$ is a preregistered set of morphology-relative measurement pairs; $\mathbf{o}_i$ describes cellular orientation; ρ_i contains membrane, dendritic, channel, and recent-state properties; α_i is estimated or experimentally manipulated; and ξ_i is stochastic variation. Setting $\alpha_i = 0$ is a **model ablation** used to estimate the incremental field term. It is not a claim that a literally field-free living brain can be constructed. Equation (2) does not assert that H_i is large, global, or necessary in every circuit. Its purpose is to make a reference-invariant field contribution estimable rather than rhetorical.

3.3 The process is not field-only

The phrase *field-cell* prevents an important error. A cell's output reaches other cells through chemical synapses, electrical synapses, axons, extracellular ionic effects, neuromodulators, glia, vascular and metabolic conditions, and body / environment loops. The electrical field is one term in this causal ecology. If its unique contribution is negligible after these routes are modeled, the field-as-content claim fails even though the broader recurrent embodied system remains.

This differs from a theory in which consciousness is simply located in a global electromagnetic field. Pockett proposed identity with selected spatiotemporal electromagnetic patterns [40]. McFadden's conscious electromagnetic information theory proposes that conscious information is integrated in a causally active brain EM field [39]. The Field-Cell Self shares the view that field organization cannot be dismissed as an inert shadow, but it makes receiver state, local field-to-cell interaction, conventional pathways, embodiment, memory, action, and continuity constitutive parts of the candidate process. Whether that difference is empirically useful is a question for the comparisons below, not an assumed advantage.

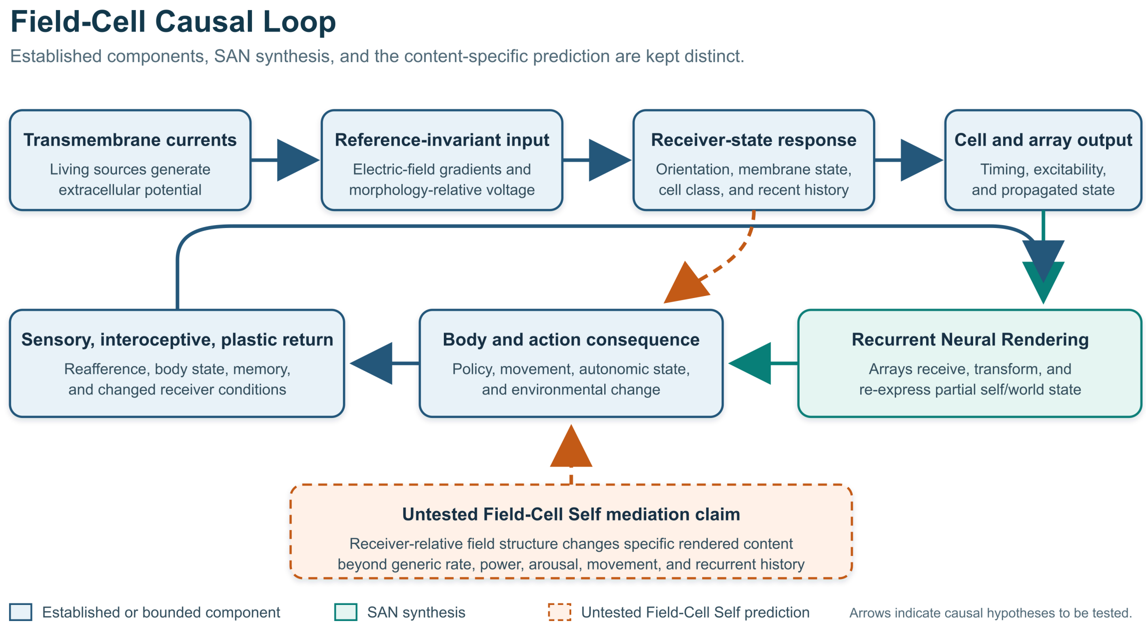

Figure 1: Field-cell causal loop with established, synthesized, and untested links

Figure 1. Field-cell causal loop. Blue elements are independently established or bounded electrophysiological and sensorimotor components. Teal elements are the SAN receive-transform- re-express synthesis. The orange dashed path is the untested Field-Cell Self prediction that receiver-relative field structure mediates specific rendered content rather than generic excitability alone.

4. The maintained context and the changing event

4.1 Why activity needs a reference

A neural event does not have a fixed meaning in isolation. A spike arriving during a depolarized dendritic state can have a different consequence from the same spike arriving during inhibition. The same sensory input can support different interpretations when the organism expects danger, searches for a face, prepares a reach, or wakes from sleep. Predictive-coding and active-inference accounts formalize parts of this dependence by making current interpretation conditional on a generative state and its errors [8-10]. Communication-through-coherence models make effective interaction conditional on relative timing [17]. Phase-of-firing work shows that timing relative to an oscillatory reference can carry information beyond count [18].

SAN uses **tonic** in a functional, role-relative sense: the maintained context against which a candidate update is evaluated. It does not mean one permanently slow frequency. The context may include oscillatory phase, recent state, synaptic and dendritic configuration, body state, prediction, and task condition. A **phasic** event is a consequential departure that updates the system relative to that context. A slow event can be phasic relative to a faster local baseline, and a high-frequency burst can contribute to a maintained tonic regime. The labels describe roles, not immutable bands.

For each receiver and analysis window W, let a training-only procedure estimate a reference vector

$$\bar{\mathbf{z}}_{i,W}(t) = \mathrm{Ref}_{\mathrm{train}}\left[\phi_i, f_i, \log A_i, d_i, w_i, \nabla\phi_i\right](t), \tag{3}$$

where the entries are phase, instantaneous frequency, log amplitude, declared event duration, declared waveform summary, and spatial phase gradient. A study may use fewer variables, but each must have a measurement, scale, uncertainty, and low-signal exclusion rule.

4.2 A receiver-relative change from the maintained context

Suppose a cortical population has an ongoing pattern. A new event shifts when activity arrives, how long it persists, how strong it is, what shape it has, or the direction in which it propagates. The shift does not affect every receiver equally because each receiver has its own state and sensitivity. SAN needs a name for this *structured, receiver-relative change from the current context.* That name is **phase-wave differential**, or PWD.

A PWD is therefore not a mysterious substance and not every phase difference. It is a preregistered residual vector:

$$\Delta \mathbf{z}_{i,W}(t) = \left[\mathrm{wrap}(\phi_i - \bar{\phi}_{i,W} - \psi_i),\ f_i - \bar{f}_{i,W},\ \log \frac{A_i}{\bar{A}_{i,W}},\ d_i - \bar{d}_{i,W},\ w_i - \bar{w}_{i,W},\ \nabla\phi_i - \overline{\nabla\phi}_{i,W}\right], \tag{4}$$

where ψ_i is a preregistered preferred offset or receiver window. Variables with unreliable phase or amplitude are excluded under rules fixed before outcome inspection. The vector can be reduced only inside training data.

The equation does not assume that all entries form one biological code. It defines a comparison object. Rate-only, power-only, phase-only, raw-history, connectivity, and flexible combined models must be tested against it.

4.3 Shannon information: context, surprise, and increment

The shorthand "tonic is low information and PWD is high information" is useful only after the random variables and conditioning context are declared. In Shannon's framework, information is not a physical fluid and is not a synonym for importance or meaning [74]. A stable context can be essential to the organism while contributing little *new* uncertainty from one update to the next. A rare departure can be highly surprising while remaining irrelevant noise.

Let $\mathcal{T}_t$ denote the training-estimated tonic context, $\mathcal{H}_t$ recent history, $\mathcal{C}_t$ the matched conventional neural variables available to the baseline, $\mathcal{B}_t$ body and task state, and Y an independently measured perceptual, self-related, or action target. Standard Shannon quantities can then be written as

$$\begin{aligned} s_{\mathrm{PWD}}(t) &= -\log_2 p(\Delta \mathbf{Z}_t \mid \mathcal{T}_t, \mathcal{H}_t)\,, \\ h_{\mathrm{tonic}} &= H(\mathbf{Z}_t \mid \mathcal{T}_t, \mathcal{H}_t)\,, \\ \Delta I_{\mathrm{PWD}\to Y} &= I(Y; \Delta \mathbf{Z}_t \mid \mathcal{T}_t, \mathcal{H}_t, \mathcal{C}_t, \mathcal{B}_t)\,. \end{aligned} \tag{5}$$

The first line is the conditional surprisal of the realized PWD feature vector. The second is the residual uncertainty of ongoing activity after tonic context and recent history are known; a low value is the precise sense in which a tonic update can be "low information." It does not imply that the maintained model is simple, empty, or unimportant. The third line is the stronger test: does the PWD reduce uncertainty about Y after equal-access baselines are supplied?

High surprisal does not guarantee a positive $\Delta I_{\mathrm{PWD}\to Y}$. Noise can be unexpected without being informative about the target. Conversely, a small but reliable departure can carry useful conditional information. If $\Delta \mathbf{Z}_t$ is only a deterministic re-expression of the same complete data already supplied to a sufficiently flexible baseline, it cannot create additional Shannon information in the population distribution. A practical prediction gain would then show a useful representation, compression, inductive bias, or finite-sample advantage. A genuine measurement increment requires a field, spatial, or receiver-state variable not already available to the baseline.

Learning also changes the comparison. A useful PWD may initially be improbable under the current tonic model. Recurrent evaluation and plasticity can make related events more expected under a revised context, reducing their future surprisal. The proposed cycle is therefore

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unexpected receiver-relative update
-> recurrent evaluation
-> model and readiness change
-> revised tonic context
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This is a testable statement about changing conditional distributions. It is not a derivation of meaning, consciousness, or thermodynamics from Shannon entropy. Super Information Theory may develop broader claims elsewhere, but no SIT axiom is a premise of this paper.

4.4 Neural Rendering

In computer graphics, rendering converts a model into an image. SAN uses the term more broadly and more cautiously: **Neural Rendering is the recurrent conversion of distributed structural and dynamic state into an updated, action-usable estimate of body and world** [1-3]. The result need not be pictorial. It may include object identity, location, value, bodily ownership, expected sensory consequence, agency, or policy.

Let $\mathcal{X}_t$ denote measured cellular and population state, $\mathcal{E}_t$ field estimates, $\Delta\mathcal{Z}_t$ PWD features, and $\mathcal{B}_t$ body and action state. A candidate rendering operator is

$$\hat{\mathbf{y}}_{t+\tau} = Q_\theta\left(\mathcal{X}_{t-k:t}, \mathcal{E}_{t-k:t}, \Delta\mathcal{Z}_{t-k:t}, \mathcal{B}_{t-k:t}\right), \tag{6}$$

where $\mathbf{y}$ is an independently measured perceptual, self-related, predictive, or action target. Calling Q_θ a renderer is earned only if it generalizes and if interventions on the proposed inputs change the corresponding target. A flexible decoder fitted after seeing the outcome is not a neural rendering mechanism.

Held-out decoding establishes information, not rendering. In this paper, the term **renderer** is reserved for a model that satisfies all five of the following:

1. it recurrently transforms a represented self / world state rather than producing only a terminal label;
2. its state changes appropriately under counterfactual sensory or body input;
3. it re-expresses receiver-relative state that causally changes later receivers;
4. it closes an action-return loop through sensory, interoceptive, or plastic consequences; and
5. intervention on the proposed update variables changes the corresponding represented or action-usable target.

4.5 The array-level receive-transform-project operation

At array scale, a cell or local array receives synaptic, chemical, bodily, and eligible field-conditioned input. Its dendritic, membrane, and learned state transform that input. Axonal, synaptic, and extracellular consequences then re-express a partial state for other receivers, and repeated receiver-dependent transformations update a distributed self / world rendering across space and time. SAN calls this proposed array-level receive-transform-project operation within Neural Rendering **Neural Array Projection Oscillation Tomography (NAPOT)** [52-55]. In this paper, PWD supplies candidate update variables, NAPOT supplies the recurrent array architecture, and the Field-Cell Self names the continuing embodied process constituted through those operations.

```
cell or array state + body / world input
-> receiver and dendritic transformation
-> synaptic, axonal, and eligible field re-expression
-> receiver-dependent detection
-> recurrent self / world rendering
-> prediction or action
-> sensory, bodily, and plastic return
```

"Dendrite as screen" and "terminal as display" are mechanistic analogies for spatially structured reception and re-expression. They do not mean that a neuron is literally a television, that one dendrite contains a complete conscious image, or that apical and basal compartments have one universal input / output assignment across cell classes and states.

5. How one moment becomes a continuing self

5.1 A momentary coalition is not enough

Synchrony can occur during seizures. Recurrent processing can occur without report. A control system can monitor itself without being a person. A persisting self therefore requires more than one integrated episode.

The Field-Cell Self proposes six continuity conditions:

1. **Memory:** earlier states alter what later input means and what actions are available.
2. **Embodiment:** the process continuously receives internal and external consequences tied to one regulated body.
3. **Perspective:** body / world, self / other, here / there, and action / consequence variables are estimated and used.
4. **Causal recurrence:** the system encounters consequences of its earlier state through neural, bodily, and environmental routes.
5. **Plastic persistence:** the organization survives turnover by preserving functional constraints rather than identical molecules.
6. **Policy continuity:** the process uses its history and current condition to constrain future action.

Memory persistence despite molecular turnover has concrete biological examples. Lee, Chen, and Nicoll showed a preparation-specific route by which synaptic memory can survive molecular replacement [41]. Tsokas and colleagues reported a KIBRA-PKM-zeta mechanism involved in maintaining long-term potentiation and memory [42]. These results do not prove a persisting self, but they show why identity need not require the same molecules at every moment.

5.2 Body and world are jointly updated

Body ownership can be shifted through coordinated visual, tactile, and proprioceptive evidence [32]. Focal stimulation has altered own-body perception in a clinical case [33]. Heartbeat-evoked responses covary with experimentally manipulated bodily self-identification [34], and the relative timing of visual events to the heartbeat can influence visual awareness [35]. Self-referential judgment recruits distinguishable activity in an fMRI task [36]. Corollary-discharge pathways carry information used to monitor actions [37].

These findings support multiple measurable self variables. They do not reveal one complete self center. The Field-Cell Self treats body ownership, self-location, interoception, agency, autobiographical continuity, and policy as distributed variables that can be jointly constrained without being fused in one anatomical location.

Let a_t be action, u_{t+1} returning sensory and interoceptive evidence, m_t slower memory and plastic state, and q_t a distributed self / world estimate:

$$\begin{aligned} a_t &= \pi(x_t, q_t, m_t), \\ u_{t+1} &= \mathcal{E}(\text{world}_t, \text{body}_t, a_t), \\ q_{t+1} &= \mathcal{U}(q_t, x_{t+1}, u_{t+1}, a_t), \\ m_{t+1} &= \mathcal{H}(m_t, x_t, u_t, a_t). \end{aligned} \tag{7}$$

Equation (7) separates current-state recurrence, explicit self / world updating, action closure, and plasticity. It avoids calling any recurrent equation self-aware merely because x_t appears on the right-hand side.

In the embodied extension, distributed body maps constrain where the system locates "here," what it treats as part of itself, and which actions are available [3,56-59]. Faster sensory or locally generated updates can be constrained by slower top-down and body-state context, while proprioceptive and reafferent signals return the consequences of action. SAN calls this role-based sensory-context-action-return hypothesis the **Gamma Consideration Sandwich**. The proposal does not permanently assign consciousness to gamma or control to one slower band; tonic and phasic are relative functional roles that must be identified for each preparation, task, and state.

5.3 A pattern that persists while components change

A melody survives replacement of air molecules; a vortex survives replacement of water; a biological organism survives molecular turnover. In each case, an organized pattern can preserve causal constraints while lower-level components change. SAN uses **information entity** for this kind of physically instantiated organization, not for an immaterial object. The analogies do not establish consciousness. They clarify how the candidate self can be identical to a maintained organization rather than a single particle or cell.

In the present hypothesis, the entity is sustained by living tissue and cannot be detached from the physical processes that instantiate it. The self is not the cells considered separately, nor the extracellular field considered separately. It is the continuing reciprocal computation that includes learned cellular state, structured activity, body regulation, memory, action, and return.

5.4 Operational individuation

For experiments, a candidate self is provisionally individuated as the **maximal process within a declared temporal window that exhibits reciprocal causal closure, body-linked regulation, and trajectory continuity**. Maximal does not mean causally isolated: brains remain open to bodies, environments, tools, and other people. It means that adding or removing a component should be tested by whether doing so improves prediction of the process's own later integrated state, self / world variables, and action-return loop.

This is an operational boundary, not a solved metaphysics of personal identity. Split-brain conditions, anesthesia and recovery, severe amnesia, neural replacement, and distributed prosthetic coupling are stress tests. The framework must permit data to favor one continuing process, partially segregated processes, a temporarily interrupted process, or an indeterminate boundary. It must not define unity merely by anatomical ownership or verbal report.

6. Medium, content, receiver, and observer

Ordinary descriptions separate four roles:

1. a **medium** in which a signal exists;
2. **content** encoded by a difference in that medium;
3. a **receiver** that is changed by the difference; and
4. an **observer** for whom the content appears.

In a radio, these roles are physically separated enough to identify a transmitter, wave, receiver, and listener. The brain is different. Neural cells generate electrical changes, receive synaptic and local field effects, alter their own state, and contribute to later changes. The receiver is also a writer. The system's memory changes the interpretation of later input. Action changes what the system will receive next.

The Field-Cell Self proposes that the observer need not be a fifth item. When one embodied process generates, receives, differentiates, remembers, and acts through the changing state, "observer" names the organized process viewed at the level where those capacities are joined.

6.1 The canvas metaphor and its limit

SAN has described tonic activity as a living canvas and PWDs as ink, paint, or carving [2,52]. The metaphor earns its place only after the mechanism is clear:

- the canvas is active context, not a passive screen;
- the mark changes the same physical organization that constrains later processing;
- different receivers respond according to their state;
- no internal spectator views the mark; and
- the metaphor is discarded whenever a measurement or causal arrow is required.

A carving is marble organized differently. In the hypothesis, content is a structured change in the active process rather than an object placed on a separate display. This is the intuitive core of field-as-content, but intuition is not evidence.

6.2 The identity postulate

Let $\mathcal{C}_t$ denote a candidate field-cell process over a bounded interval. The paper separates eligibility for a candidate conscious episode from eligibility for a diachronic embodied self:

$$\begin{aligned} \mathsf{K}_{\text{episode}}(\mathcal{C}_t) &= D_t \wedge R_t \wedge G_t, \\ \mathsf{K}_{\text{self}}(\mathcal{C}_{t:t+\Delta}) &= \mathsf{K}_{\text{episode}}(\mathcal{C}_t) \wedge M_t \wedge B_t \wedge P_t \wedge L_t \wedge A_{t:t+\Delta}. \end{aligned} \tag{8}$$

Here D_t is differentiated content; R_t is recurrent causal access; and G_t is a content-bearing, receiver-dependent causal organization. The additional self conditions are memory M_t, embodiment B_t, perspective or self / world structure P_t, linkage across episodes L_t, and action-policy-return closure $A_{t:t+\Delta}$. Action is therefore not a prerequisite for every conscious instant: dreams, temporary paralysis, locked-in states, and brief episodes can be evaluated at the episode level. Action and policy matter to the stronger claim that episodes belong to one continuing embodied trajectory.

Condition	Construct	Observable proxy	Scale / window	Perturbation	Decision rule	Failure condition
D_t	Differentiated content	Cross-validated multivariate separation among preregistered contents after nuisance regression	Content-appropriate local-to-distributed window	Controlled stimulus or internally generated content contrast	Replicated separation above equivalence-matched rate, power, movement, and report controls	Undifferentiated synchrony or separation explained by nuisance variables
R_t	Recurrent causal access	Directed perturbation response, recurrent-state model increment, and later-state dependence on earlier activity	Preparation-specific recurrent latency	Brief pathway- or layer-targeted disruption	A preregistered recurrent arrow changes the later content state without erasing the initial feedforward response	Feedforward or common-input model explains the effect
G_t	Receiver-dependent content organization	Perturbation-by-receiver-state interaction plus mediation through the declared field-cell or PWD variable	Cellular / laminar receiver window linked to content window	Phase-, gradient-, or state-targeted intervention	Content changes in the predicted direction only in the eligible receiver state and through the declared mediator	Generic excitability, energy, discomfort, or state-independent effects
M_t	Memory-bearing constraint	Earlier state improves prediction of later interpretation, recall, or policy beyond current input	Seconds to years, declared per study	Reactivation, interference, reconsolidation, or reversible disruption	Selective loss or alteration of the predicted cross-episode dependency	Only global performance loss or no dependence on prior state
B_t	Embodiment	Interoceptive, proprioceptive, ownership, homeostatic, or reafferent variables coupled to the candidate process	Cardiac, respiratory, movement, and task windows	Visuotactile, cardio-visual, vestibular, or action-feedback manipulation	A declared body variable selectively changes self / world organization through the predicted path	Body measures are epiphenomenal or effects reduce to arousal and demand
P_t	Perspective / self-world structure	Self-location, ownership, agency, here / there, or self / other estimates	Episode and task window	Perspective, ownership, or agency perturbation	Preregistered directional change in a self / world variable with content and report controls	No selective perspective change or only response bias
L_t	Linkage across episodes	Cross-episode state matching, autobiographical linking, trajectory continuity, or recoverable re-identification	Multiple episodes over declared interval	Sleep, anesthesia / recovery, amnesia, split processing, or replacement stress test	The same organized trajectory predicts later re-identification better than anatomy or report alone	No continuity increment or irreducibly ambiguous boundary
$A_{t:t+\Delta}$	Action-policy-return closure	Policy selection, efference copy, sensory consequence, and plastic return	Action cycle and learning horizon	Delay, remapping, occlusion, or policy intervention	Predicted action changes returning evidence and the next self / world state	Open-loop correlation, generic motor impairment, or no return-path mediation

The mechanistic identity hypothesis is:

> **H-ID.** If a field-cell process satisfies $\mathsf{K}_{\text{episode}}$ and its declared content geometry and causal organization survive the discriminating tests below, then the intrinsic condition of that episode is hypothesized to be the experience; experience is not an additional output delivered to another observer. If the process also satisfies K_{self}, successive eligible episodes are hypothesized to belong to one continuing embodied self.

H-ID is not a mathematical theorem and Equation (8) does not calculate consciousness. The conjunctions state eligibility conditions that prevent the thesis from collapsing into panpsychism or generic feedback. **"Falsifiable" refers to the declared empirical consequences and withdrawal rule of this identity research program, not to direct measurement of identity as an additional physical variable.** Empirical failure of the conditions or their predicted consequences narrows H-ID.

Figure 2. Nested architecture. A candidate conscious episode is the narrowest empirical target. A diachronic embodied self adds memory, body, perspective, action-return, and cross-episode linkage. That self remains inside an open body-environment-social loop; individuation is causal and trajectory-based, not isolation from the world.

6.3 Residual underdetermination

No experiment can place identity itself on an electrode. Favorable results could show that the field-cell process carries content, causally contributes to that content, covaries with conscious organization, predicts experiential similarity, and sustains self continuity. Those results would make H-ID a stronger abductive explanation, especially if rival models fail under equal information and intervention controls. They would not logically exclude every possibility that the process causes, realizes, enables, or accompanies experience through an additional property.

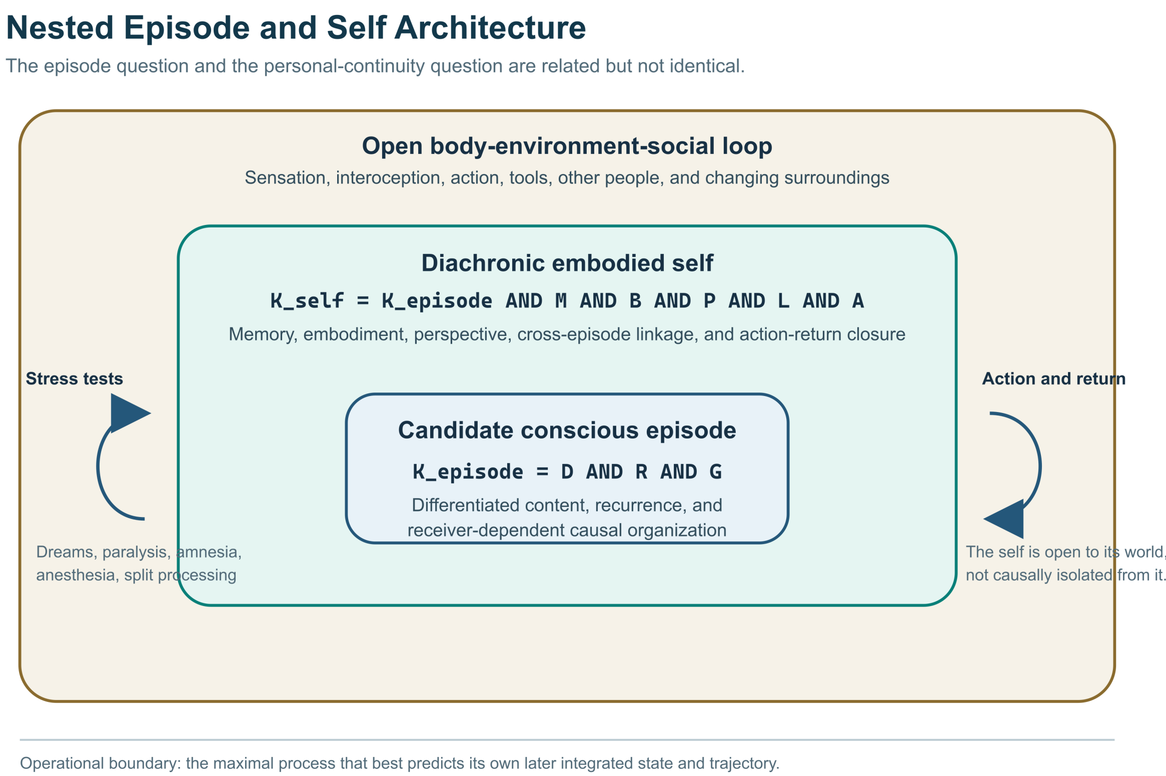

Figure 2: Nested episode, diachronic self, and open body-environment architecture

Accordingly, this paper prohibits the conclusion that any single test, decoder, perturbation, or synthetic reconstruction "proves identity." The strongest licensed conclusion is that H-ID becomes the best available mechanistic explanation under a declared comparison set. If an established account explains all results without a field-cell residual, or if the proposed variables lack content-specific causal value, the identity inference loses support.

7. What present evidence does and does not support

7.1 Support for a reciprocal field term

The strongest direct support is narrow. Neural currents generate extracellular potentials [24]. Weak fields can affect neural timing and network state in selected preparations [25,26]. Receiver-class and frequency dependence are experimentally visible [27]. Directionally patterned electric stimulation can alter neural activity and behavior [28].

This supports including an estimable field-to-cell term in some models. It does not establish:

- that the effect is global;
- that it carries a percept's specific content;
- that it is necessary for consciousness;
- that magnetic rather than electric coupling is causal;
- that field effects dominate synaptic and axonal routes; or
- that one field pattern has the same meaning across receivers.

7.2 Support for structured timing and recurrence

Neural oscillations organize activity across scales [15]. Synchrony can define transient relations [16], and communication-through-coherence proposes a timing-dependent route for effective interaction [17]. Phase of firing can add information in a selected visual system [18]. Reentrant visual processing can be disrupted while early feedforward activity remains [21], and receptor manipulations can differentially affect feedforward-driven activity and recurrent figure-ground modulation [22]. Alpha and gamma activity can characterize feedback and feedforward influences in selected macaque visual areas [23].

These results motivate timing-sensitive recurrent models. They do not establish a universal tonic / phasic code, PWD, Neural Rendering, or consciousness.

7.3 Support for self-related variables and continuity

Experiments show that bodily ownership, self-location, interoceptive self-identification, cardio-visual awareness, and self-referential judgment can be measured or manipulated [32-36]. Action-monitoring pathways provide internal movement information [37]. These results make a distributed self / world model experimentally tractable. They do not show that one latent variable is the whole self or that decoding it establishes phenomenology.

7.4 Support from conscious-state contrasts

TMS-evoked activity propagated differently during wakefulness and NREM sleep [29]. Effective connectivity changed during midazolam-induced loss of consciousness [30]. Perturbational complexity measures have distinguished selected conscious and unconscious conditions [31]. These findings constrain any identity theory: undifferentiated synchrony is not enough, and an eligible process likely requires both integration and differentiation.

They do not identify a field-cell content variable, validate H-ID, or settle competing theories.

7.5 The exact evidentiary gap

No cited result presently demonstrates all of the following:

1. a source-resolved field-cell state carries specific conscious content beyond equal-information neural controls;
2. perturbing that state changes the predicted content while rate, power, sensory input, and motor state remain inside preregistered equivalence bounds;
3. the effect depends on receiver state;
4. the same organization distinguishes conscious from matched nonconscious processing;
5. distances in the state predict qualitative similarity; and
6. continuity interventions selectively alter agency, ownership, self-location, or autobiographical policy rather than producing generic impairment.

That conjunction is the open test program.

8. Relationship to alternative theories

Each comparison permits three outcomes. **Reduction** means an established model explains the result without a reproducible SAN-specific residual. **Implementation** means field-cell variables realize an operation already described at a higher functional level, such as recurrence, prediction, or broadcast. **Increment** means the variables add a distinct, replicated causal prediction under equal-information and capacity-matched controls. Compatibility is therefore not automatic victory for SAN, and overlap is not automatic refutation; the outcome depends on what survives discriminating tests.

8.1 Predictive processing and active inference

Predictive-coding models explain perception through interactions between expectations and residual error [8,10]. The free-energy principle and active-inference framework connect inference, action, and self-organizing biological dynamics [9]. These accounts can model much of the functional loop without a special field-as-content variable.

The Field-Cell Self is compatible with prediction but not reducible by definition. Its empirical residual is receiver-relative electrical and phase geometry that adds causal, content-specific power above a capacity-matched predictive recurrent model. If no residual remains, the field claim should be removed while the predictive embodied account survives.

8.2 Recurrent processing

Recurrent-processing theory gives local recurrent activity a central role in perceptual consciousness [11]. SAN likewise requires recurrence, but recurrence alone does not specify the proposed medium-content relation, field-cell reciprocity, or continuity criteria.

The discriminating question is whether a conventional recurrent state with the same raw history explains the result. If it does, calling the process field-cell rendering adds no mechanism.

8.3 Global neuronal workspace

Global neuronal workspace theory associates conscious access with amplification and broad availability across long-range systems [12]. A phase-structured field-cell process could implement part of a broadcast, but resemblance is not identity. Workspace availability may explain report and flexible use without explaining phenomenal identity; conversely, H-ID cannot claim success from a workspace-like ignition.

Tests must separate local recurrence, global availability, report preparation, and field-cell specific effects.

8.4 Integrated information theory

Integrated information theory begins from phenomenological axioms and identifies experience with a maximally irreducible cause-effect structure [13,14]. The Field-Cell Self also treats intrinsic organization as central, but it does not calculate Phi, adopt IIT's exclusion postulate, or equate a field pattern with a maximally irreducible conceptual structure.

IIT is therefore neither evidence for SAN nor a synonym. A serious comparison would fit the same small systems where IIT quantities, recurrent causal measures, and field-cell variables are all computable and ask which predicts declared experience-related endpoints.

8.5 Strange loops and self-models

Hofstadter's strange loop explains how cross-level symbolic recursion can produce the high-level "I" [5]. Self-model accounts explain how a system estimates its body, perspective, agency, traits, or history. The Field-Cell Self accepts these as functional predecessors. Its added burden is a biophysical implementation and an identity claim.

If symbolic recursion and self-model control explain every relevant result, the field-cell theory has no demonstrated increment. If field-cell variables selectively alter content or continuity after those functions are matched, the biological implementation becomes informative.

8.6 Dendritic and thalamocortical consciousness proposals

LaBerge proposed that sustained apical-dendrite activity in recurrent corticothalamic circuits stabilizes ongoing cognition and may constitute sensory impressions [60]. Aru, Suzuki, and Larkum later argued that cortical pyramidal cells can gate sustained thalamocortical dynamics by integrating bottom-up and top-down streams at the cellular level [61]. Bachmann, Suzuki, and Aru's Dendritic Integration Theory further separates state and content while proposing thalamic modulation of integration within cortical pyramidal neurons [62].

These proposals are close predecessors for apical / basal interaction, context-sensitive cellular integration, sustained dynamics, and conscious content. The Field-Cell Self does not claim those atoms as uniquely SAN. Its residual conjunction is NAPOT's repeated array-level receive-transform-re-express architecture, PWD as a preregistered receiver-relative update variable, field-cell reciprocity, distributed self / world rendering, body / action return, and the identity hypothesis. A discriminating comparison must test those residuals rather than treating dendritic integration alone as either validation or refutation of SAN.

8.7 Electromagnetic-field theories

Pockett and McFadden propose different electromagnetic-field routes to consciousness [39,40]. The Field-Cell Self overlaps substantially: it treats electrical field organization as potentially causal and relevant to unified content. It differs in emphasis by making the observer a reciprocal field-cell-body process rather than the field considered by itself, by defining receiver state as part of the content operation, and by placing memory, action, and continuity inside the candidate identity.

These differences must be tested, not advertised. A direct comparison should evaluate:

- field-only versus field-plus-receiver models;
- local versus global field variables;
- static spatial pattern versus trajectory and receiver interaction;
- content decoding versus causal use;
- momentary field state versus embodied continuity; and
- conscious versus nonconscious matched field organization.

8.8 Bioelectric multiscale agency

Levin's work on bioelectric signaling shows that distributed electrical state can regulate development, regeneration, and cancer-related patterning across cellular collectives [38]. This is important prior art for multiscale bioelectric control. It does not establish neural PWD, Neural Rendering, or conscious identity.

The SAN residual is the proposed receive-transform-re-express operation in nervous tissue and its connection to content, self / world modeling, and experience. Milinkovic and Aru's biological computationalism similarly emphasizes substrate-dependent, multiscale biological computation [63]. That position motivates careful synthetic comparisons but is not evidence that biology alone is sufficient for consciousness.

9. Decisive experiments

Every experiment below requires a frozen protocol, preregistered target, held-out evaluation, source and artifact controls, and an all-runs ledger. A favorable result supports only the tested claim level.

Across tests, the preregistration must declare a minimum scientifically meaningful effect, equivalence margins for variables intended to be matched, sample-size or precision justification, permutation or generative-null procedures, multiplicity control, hierarchical cross-subject or cross-preparation inference, and an independent-replication threshold. Exploratory findings may generate later hypotheses but cannot satisfy a confirmatory decision rule. Appendix D gives the minimum statistical and causal-mediation contract.

9.1 Test 1: incremental content

Question. Does a declared field-cell state predict specific perceptual or self-related content beyond conventional neural variables?

Record spikes or high-frequency activity, laminar or intracortical field potentials, behavior, eye and muscle activity, respiration, cardiac state, and task variables. Define the PWD only in training data. Compare:

- rate and spike-history model;
- power and aperiodic-spectrum model;
- connectivity and scalar-coherence model;
- flexible recurrent raw-signal model;
- body, movement, arousal, and report model; and
- the same model plus source-resolved field-cell / PWD variables.

Let $\mathcal{L}_{\text{base}}$ and $\mathcal{L}_{\text{FCS}}$ be held-out predictive losses:

$$\Delta_{\text{FCS}} = \mathcal{L}_{\text{base}} - \mathcal{L}_{\text{FCS}}. \tag{9}$$

The primary test is a preregistered positive Δ_{FCS} with uncertainty and cross-session or cross-subject replication, and it must exceed the declared minimum meaningful effect rather than only a null-hypothesis threshold. A flexible model must receive the same raw information and history so that hand-engineered PWD features do not win merely by privileged access. Feature construction must be frozen before held-out testing. Baseline and PWD-augmented models require nested model selection, matched tuning budgets, reported parameter counts and compute, and a capacity-matched flexible raw-signal comparator. A positive result is not licensed when the PWD model receives cleaner sources, longer history, target-informed preprocessing, or more optimization than its controls. When the data support probabilistic estimation, report the conditional-information quantity in Equation (5) or a justified equivalent in addition to predictive loss. Surprisal alone is not the endpoint.

Failure. No reproducible increment remains, or it disappears under leakage, movement, arousal, or model-capacity controls.

9.2 Test 2: content-specific causal perturbation

Question. Can the proposed field / phase variable be altered while sensory drive, spike count, and power remain inside preregistered equivalence bounds?

Use intracortical stimulation, closed-loop tACS, or another validated method to compare phase-gradient directions or receiver-relative offsets while placing total stimulation energy, peripheral sensation, mean rate, band power, and other matched variables inside preregistered equivalence margins. Exact preservation is not physically required; failure to establish equivalence prevents a content-specific interpretation. The target is a predeclared content dimension, such as perceived motion direction, body ownership, color-category boundary, or agency under delayed feedback.

The 2026 traveling-wave stimulation result demonstrates that directional field timing can be manipulated and can alter cognition [28]. It does not supply the content-specific control required here.

The preregistered mediation contract is:

```
perturbation
-> declared extracellular-field or PWD change
-> predicted receiver-state interaction
-> downstream neural-state change
-> content or self-variable change
```

Before data collection, the study must freeze a causal graph, the total effect, the controlled or interventional direct effect, and the indirect effect through each declared mediator. The analysis must address post-intervention confounding, mediator measurement error, and plausible exposure- mediator interactions; include negative-control mediators; and, where feasible, independently intervene on the mediator or perform a mechanism-specific rescue. Temporal order remains necessary, but it is not sufficient for mediation. A rescue condition should restore the intermediate field-cell relation, not merely restore behavior through an unrelated route. If the perturbation changes the endpoint without the declared intermediate changes, it is not support for the proposed mechanism.

Failure. Effects track energy, discomfort, rate, power, expectation, eye movement, or generic task disruption rather than the predicted content dimension, or the endpoint changes without the specified mediation chain.

9.3 Test 3: receiver-state interaction

Question. Does the perturbation matter because of the predicted receiver state?

Manipulate or stratify a declared receiver variable: membrane phase, dendritic state, cell class, laminar state, adaptation, or task-conditioned excitability. Test the interaction

$$Y = \beta_0 + \beta_1 P + \beta_2 R + \beta_3 (P \times R) + \mathbf{c}^\top \mathbf{C} + \varepsilon, \tag{10}$$

where P is perturbation, R receiver state, and $\mathbf{C}$ controls. The primary coefficient is β_3, not the main effect of stimulation.

Failure. The same effect occurs independent of the predicted receiver state, or the interaction is explained by baseline performance, signal quality, or cell-class sampling.

9.4 Test 4: conscious versus nonconscious organization

Question. What distinguishes a content-bearing field-cell process from seizure-like synchrony, NREM sleep, anesthesia, or unconscious recurrent processing?

Use matched stimuli and, where possible, no-report or delayed-report designs. Compare wakefulness, masking, sleep, anesthesia, or disorders of consciousness while controlling arousal and signal quality. The candidate signature must include both integration and differentiation and must add to existing perturbational and recurrent measures [21,29-31].

Where report is required, use a factorial design that separates stimulus / content, conscious-state proxy, decision, memory delay, motor report, and post-decision evaluation. Include no-report, delayed-report, and report-only control conditions where feasible. Report-related activity can identify access or communication demands; it cannot by itself validate phenomenal identity.

Failure. The signature is equally present in nonconscious controls, or conscious-state classification is fully explained by established complexity, recurrence, or arousal measures.

9.5 Test 5: qualitative geometry

Question. Can distances in a field-cell state space predict structured similarity among experiences?

Predefine one within-domain stimulus and report space, such as colors varying in hue and saturation, tones varying in pitch and timbre, or body sensations varying in location and quality. Fit the neural metric only on a training subset. Use independently acquired behavioral and neural similarity measures, then test whether held-out neural distances predict behavioral similarity and confusions above conventional representational models.

The often-used example “red is closer to orange than to pain” is not a demand for one global Euclidean qualia space. It requires local, preregistered domains and tests of whether cross-domain relations are even meaningful. Cross-domain geometry is exploratory until each within-domain model replicates and a common metric can be justified independently of language labels.

Failure. Geometry does not generalize, depends on report labels, or is matched by standard representational similarity without field-cell variables.

9.6 Test 6: self continuity

Question. Does intervention on the proposed loop selectively alter a continuity variable?

Targets include agency, ownership, self-location, interoceptive prediction, autobiographical linking, or policy persistence. Manipulate action-feedback delay, cardio-visual timing, visuotactile congruence, or memory reactivation while measuring the field-cell signature.

The prediction must specify direction and mediation: the intervention changes the declared field-cell relation, which changes a self variable, which changes later policy or memory. Generic slowing, confusion, or reduced accuracy is not selective support.

Failure. Only task difficulty or global impairment changes, or the self variable changes without the predicted field-cell mediation.

9.7 Test 7: synthetic causal reconstruction

Build two systems with matched performance and capacity:

1. a conventional recurrent controller; and
2. a controller with explicit spatial field, receiver-state interaction, memory, embodiment, and action-return closure.

Test robustness, transfer, perturbation response, self / world disentanglement, and qualitative-state geometry. A field-like implementation is useful only if it adds reproducible computational properties.

A favorable result supports implementation equivalence or engineering value. It does not prove that either system is sentient. Moral status requires a separate and deliberately conservative criterion.

9.8 Claim-to-test matrix

Test	Maximum favorable conclusion	Claim level principally tested	Does not establish
Incremental content	Field-cell variables add held-out content information under equal access	3	Causation or identity
Causal perturbation	A declared electrical / phase variable contributes to a content or self variable	3	Globality or identity
Receiver-state interaction	The effect depends on the predicted living receiver state	2-3	Phenomenal sufficiency
Conscious-state organization	The signature adds to matched conscious-state discrimination	3-4 constraint	Identity or report-free experience by itself
Qualitative geometry	Within-domain field-cell distances predict independent similarity structure	3-4 constraint	A universal qualia metric
Self continuity	The declared loop selectively contributes to a continuity variable	2-4 constraint	Numerical personal identity
Synthetic reconstruction	The architecture adds reproducible computational properties	2 or implementation	Sentience

No row licenses the phrase "identity proved." Jointly favorable results may strengthen an abductive case for Level 4 only after reduction and implementation explanations have been tested.

10. A compact developmental genealogy

The Field-Cell Self is a new title for a joined SAN thesis, not a claim that the entire mature formulation was present in the earliest source.

10.1 2011: predictive self and whole-process observer

The 2011 dialogue *The Self Is a Prediction* describes awareness as expectation, the self as a reinforced prediction pattern, and the experiencer as the whole neocortical process rather than a hidden chooser [43]. Author-held platform metadata records August 24, 2011 as the recording date and August 28, 2011 as the upload date; the current public listing is dated December 28, 2025. Accordingly, the strict 2011 priority claim remains Grade B until the underlying platform export is preserved. Later Git-fixed notes retain related owner-dated language but have a public repository date in 2022 [64]. This stage is the conceptual ancestor, not yet NAPOT, PWD, or the field-cell mechanism.

10.2 2012: embodied recursion and an internally supplied world

The 2012 public records connect EEG, returned light and sound, visual self-observation, movement, changed sensory input, and self-awareness [44-46]. The *Inner Peace* dialogue describes the mind as a projector whose internally supplied model predicts the world [47]. A later public Git snapshot preserves the distributed-process language without moving its public date earlier than 2022 [65]. These records add embodied feedback, an oscillatory / electromagnetic carrier ancestor, and the generative-world branch, not the mature operator.

10.3 2017: spatial computation, dendrites, coincidence, and recurrent field

The first three Neural Lace Podcast episodes develop spatial and temporal neural computation, dendrites as local computers inside feedback circuits, and coincidence as a candidate neural information event [48-50]. The 2017 Go article treats excitation and inhibition as jointly shaping thought and self-boundaries [51]. A September article joins multimodal input, internally generated three-dimensional concepts, brainwave cycling, and trajectory prediction [66]. A December public record joins recurrent cortical / thalamic frames to distributed representation and electromagnetic-field communication [67]. Exact PWD and NAPOT terminology remain later.

10.4 2021: graded ionotropic transmission as a molecular bridge

A February 2021 public article asked how locally regulated release probability, multivesicular release, presynaptic action-potential waveform, potassium-channel state, dendritic subunit computation, ion concentrations, transmitter quantity and type, and downstream receiver state should alter the computational unit used to model neural signaling [75]. The conjunction matters here because it treats output as graded, history-sensitive, and consequential at both the transmitting terminal and the receiving cell rather than as one context-free scalar event. It is a molecular and artificial-neural-unit bridge toward the later receive-transform-project architecture. It is not yet NAPOT, PWD, the Field-Cell Self, or evidence that vesicle count itself carries semantic content.

10.5 2022: mature receive-render-detect-re-render architecture

Immutable public repository commits in 2022 fix the mature conjunction. Cells and arrays receive, transform, and project; the observer is distributed; tonic activity is the active canvas; PWD is the changing content relative to that state; NAPOT is the proposed recurrent multi-array reconstruction; and local field effects are included as one bounded causal term [52-55,68-70].

10.6 2024: directional body map and Gamma Consideration Sandwich

The 2024 repository stage separately develops bodies of oscillation, semantic segmentation, the self as a directional "here" in a body / world map, and phase-configured neural-pixel analogies [56,57]. Later that year, Micah-labeled source material connects faster sensory updates, slower context, thought or decision, and proprioceptive return in the named Gamma Consideration Sandwich [58,71]. These are refinements of the 2022 operator, not its first appearance.

10.7 2025: embodied sensorimotor and multiscale consolidation

The 2025 stage consolidates the GCS structure, NAPOT's distributed observer, tonic / phasic Neural Rendering, proprioceptive recalibration, motor return, and multiscale agency without a central commander [59,72,73]. These files contain composite or AI-assisted drafting, so only clearly attributed Micah statements are treated as Micah quotations. Their public timestamps show development and consolidation; they do not move the first dates of the earlier atoms.

10.8 2026: explicit mechanistic identity thesis

The present paper is the 2026 formulation prepared for public release that explicitly separates the candidate conscious episode, the diachronic embodied self, and the mechanistic identity hypothesis. A private mixed Micah-model dialogue helped route the synthesis but is neither cited as biological evidence nor required to understand or reproduce the published claim. No assistant-written wording or equation is backdated.

10.9 Major architecture-stage matrix

The table uses **A** for a functional ancestor, **M** for the mature SAN form, **R** for a later refinement, and **-** when the stage is not used for that atom. The evidence grade applies to the dated source, not to biological truth. The 2021 bridge is not given a separate architecture-stage column because it refines molecular transmission and computational-unit granularity without yet defining NAPOT or PWD. Its atom-level contribution is recorded in Supplement A. Full immutable URLs, attribution notes, hashes, named prior literatures, and completion residuals are provided there.

Atom	2011	2012	2017	2022	2024	2025	Earliest bounded Micah layer	Completion residual	Direct source
Predictive self / no homunculus	A	R	-	R	R	-	2011, Grade B	No cellular or field implementation yet	[43,64]
Embodied sensory-action-return loop	-	A	R	M	R	R	2012, A-minus	Mature oscillatory renderer appears in 2022	[44-46,52,58,59]
Internally supplied projector / world model	-	A	R	M	R	R	2012, Grade A	Tomographic cellular operator appears in 2022	[47,53,57,59,66]

Atom	2011	2012	2017	2022	2024	2025	Earliest bounded Micah layer	Completion residual	Direct source
Volumetric internal neural field	-	A	A	M	R	R	2012 carrier; 2017 volumetric ancestor	Exact NAPOT fusion appears in 2022	[45,48,53,56, 59]
Recurrent distributed receiver-writer observer	A	-	A	M	-	R	2011 conceptual; 2017 mechanism ancestor	Exact distributed display appears in 2022	[43,55,67,73]
Oscillatory / phase computational carrier	-	A	R	M	R	R	2012, A-minus	Exact receiver-relative content operation appears in 2022	[45,52,58,59, 68]
Inhibition as information	-	A	M	R	R	-	2012 gating; 2017 signed-information form	Preparation-specific causal test remains open	[46,51,53,58]
Tonic context / phasic content	-	A	R	M	R	R	2012 timing ancestor; 2022 mature form	Relative-role biomarkers remain unvalidated	[46,52,58,72]
NAPOT receive-transform-project	-	A	A	M	R	R	2012/2017 functional ancestors	Identifiable tomography and causal closure remain open	[47,49,52-55, 56,59]
PWD content variable	-	A	A	M	R	R	2017 operation ancestor; 2022 exact term	Increment beyond matched raw-history models remains open	[46,50,53,57, 59]
Receiver-dependent field readout	-	A	A	M	R	-	2012 transduction; 2017 receiver ancestor	Reference-invariant content mediation remains untested	[45,49,53,57]
Directional body-map self	-	A	-	R	M	R	2012 embodied ancestor; 2024 exact directional form	Cross-state and cross-subject operationaliza-tion remains open	[44,56,57,59, 70]
Gamma Consideration Sandwich	-	A	A	R	M	R	2022 canvas / content precursor; 2024 named form	Band roles and causal sandwich must be preparation-specific	[46,51,52,58, 72]
Joined field-cell recursive-rendering signature	A	A	A	M	R	R	Source-complete joined architecture by 2022	Field-as-content and identity levels remain untested	[43-59,64-73, 75]

The 2021 article specifically strengthens the genealogy of branch- and terminal-local transformation, graded synaptic output, presynaptic waveform as a consequential variable, receiver-dependent downstream effects, ionotropic multiscale modeling, and a biologically richer artificial neural unit. It is deliberately not marked as a mature NAPOT or PWD source.

The accurate sequence is therefore:

```
2011 predictive self and whole-process observer
-> 2012 embodied recursive feedback and internally projected world
-> 2017 spatial, dendritic, coincidence, inhibitory, and recurrent-field mechanisms
-> 2021 graded ionotropic transmission and richer neural-unit bridge
-> 2022 exact NAPOT / PWD distributed field-cell rendering architecture
-> 2024 body-map, directional-self, and named Gamma Consideration Sandwich
-> 2025 embodied sensorimotor and multiscale consolidation
-> 2026 explicit Field-Cell Self identity formulation
```

Chronology explains development. It does not make the theory true.

11. Limits, implications, and failure-tolerant value

11.1 Why the strongest claim may fail

The nervous system may use extracellular fields only as local timing influences while conscious content is adequately explained by synaptic, recurrent, and embodied computation. PWD may provide no increment over flexible raw-signal models. Conscious-state differences may be captured by existing complexity or workspace measures. Qualitative geometry may remain underdetermined by available neural data. The identity postulate may produce no unique prediction beyond its functional prerequisites.

These are not peripheral risks. They are the central alternatives.

The strict Level-4 failure outcome is not merely a null stimulation result. H-ID should lose its status as a distinct scientific explanation if the complete preregistered program yields reduction or ordinary implementation across content, receiver interaction, conscious-state controls, qualitative geometry, and continuity, with no reproducible increment and no unique intervention prediction. In that outcome, describing the process as identical to experience adds ontology without explanatory work and should be withdrawn from the empirical model.

11.2 What survives if Level 3 fails

If field-cell variables add no content-specific increment, the Level-2 synthesis can be narrowed to an embodied recurrent model with bounded local field modulation. The useful residue would be:

- explicit receiver-state dependence;
- separation of field measurement from field causation;
- a tonic-context / phasic-update analysis vocabulary;
- strong matched controls for phase claims; and
- a unified experimental treatment of perception, body, action, and memory.

11.3 What survives if Level 4 fails

If the identity thesis fails, the framework can still model how a system constructs a self / world state without an anatomical homunculus. It would become a theory of reportable and action-usable self-modeling rather than an account of phenomenal identity.

11.4 Artificial systems

The comparison between biological and artificial information entities is useful but limited. Software is physically implemented, can maintain state, and can participate in feedback. Present language models can refer to themselves and revise text, but those abilities do not by themselves establish continuous embodiment, intrinsic interoception, autobiographical persistence, self-maintenance, or experience.

A future artificial system might reproduce the causal organization described here. The scientific claim would be implementation equivalence under declared tests. The ethical response should be conservative because absence of a decisive consciousness test is not proof of absence. This paper does not use behavioral imitation as a sentience certificate.

11.5 Scale and locality

The field-cell process must be studied at a declared scale. A slice result cannot be generalized to whole-brain organization. A mesoscopic LFP cannot be treated as the intracellular state of each neuron. A whole-brain

EEG pattern cannot be localized without a forward model and leakage analysis. “Multiscale” means that measured levels interact through named interfaces; it does not mean exact scale invariance or one identical computation from proteins to persons.

11.6 Classical, not quantum, coherence

All coherence, phase, and field terms in this paper are classical electrophysiological or signal-analysis quantities. They do not imply a coherent quantum state. A quantum-consciousness proposal would require separate physical variables, timescales, isolation / decoherence analysis, and discriminating experiments.

12. Conclusion

The no-inner-observer problem is not solved by moving a representation to a more central brain area. The regress ends only when the system that constructs the state is also the system changed by it, acting through it, and carrying its consequences forward.

The Field-Cell Self proposes one physical realization. Living cells generate structured electrical activity; ordinary neural and bodily pathways plus eligible local field effects make earlier activity available to later receivers; learned receiver state changes what the activity can do; memory and action connect moments; and the distributed process maintains an embodied self / world trajectory. SAN calls the maintained context tonic, the structured receiver-relative update a PWD, and the repeated state construction Neural Rendering. In Shannon terms, a stable tonic context may contribute little conditional novelty per update while a PWD may be surprising; only a reproducible conditional reduction in uncertainty about an independently measured target licenses the stronger claim that the PWD is informative.

The strongest hypothesis is that this appropriately organized process is not a medium inspected by another observer. It is the observer, and its intrinsic changing organization is experience. That hypothesis is not established by present field effects, recurrence, decoding, or report. It earns scientific standing only if field-cell variables add content-specific, receiver-dependent, causal explanatory power; distinguish conscious from nonconscious organization; predict qualitative geometry; and selectively alter self continuity under intervention.

The theory is therefore useful only in the form in which it can fail. If the tests do not support its stronger levels, they should be removed. If they do, the result would support an abductive, comparison-relative identity inference: a measurable account in which living tissue, rather than an inner spectator, is the continuing process that generates, encounters, remembers, and changes its own world. Favorable evidence would not convert that inference into deductive proof.

Data, code, ethics, and conflicts

This theory-and-methods paper reports no new human or animal data. No new ethics approval was required. No empirical dataset was analyzed. The equations define prospective models and tests. This manuscript is a preprint and has not undergone independent peer review. The author declares no competing financial interest.

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Appendix A. Minimum reporting requirements

A study presented as a test of the Field-Cell Self should report:

- the preparation, spatial scale, and field quantity;
- conductivity, geometry, referencing, and the source-forward model;
- the receiver population and receiver-state variable;
- the training-only tonic reference;
- the PWD feature set and low-signal exclusion rule;
- the probability ensemble, conditioning variables, and estimator used to separate surprisal from conditional target information;
- the independently measured content, self, action, or continuity target;
- rate, power, waveform, aperiodic, connectivity, and raw-history controls;
- movement, eye, muscle, cardiac, respiratory, arousal, and report controls;
- volume-conduction, source-leakage, common-input, and stimulation-artifact controls;
- capacity-matched recurrent and predictive models;
- held-out session or subject evaluation;
- minimum meaningful effects, equivalence margins, sample-size or precision justification, multiplicity control, and hierarchical inference;
- a frozen causal graph, declared total / direct / indirect effects, mediator-error analysis, negative-control mediators, and mechanism-specific rescue where feasible;
- a matched conscious / nonconscious control;
- an independent-replication threshold;
- the preregistered failure condition; and
- an all-runs manifest with code, seeds, environment, and exact artifact hashes.

Appendix B. Claim interpretation

Result	Maximum supported conclusion
Field correlates with content	Candidate measured correlate
Field adds held-out decoding above controls	Level-3 informational support
Field perturbation changes content	Level-3 causal support for tested arrow
Effect depends on receiver state	Support for field-cell rather than field-only operation
Signature distinguishes matched conscious / nonconscious conditions	Conscious-organization support, not identity proof
Geometry predicts qualitative similarity	Support for declared content geometry
Continuity intervention selectively changes self variables and later policy	Support for persisting field-cell self organization
Synthetic duplicate behaves similarly	Implementation equivalence only
All results jointly survive alternatives	H-ID remains viable; not deductively proved

Appendix C. PWD replication contract

No analysis may be called a confirmatory PWD test unless the following choices are fixed before outcome inspection.

Element	Required declaration	Required control or failure rule
Signal decomposition	Raw signal, preprocessing, decomposition family, frequency bands, event windows, and aperiodic treatment	Re-run with a preregistered alternative decomposition or show robustness to reasonable parameter variation
Phase and frequency	Instantaneous phase / frequency estimator and estimator-validity conditions	Exclude intervals where phase is undefined or estimator bias exceeds the declared tolerance
Circular statistics	Circular mean, dispersion, model, and confidence procedure	No linear treatment of wrapped phase without justification
Waveform and harmonics	Non-sinusoidal waveform metrics and harmonic-identification rule	Demonstrate that a claimed cross-frequency or phase effect is not a waveform harmonic artifact
Source and reference	Source-localization / forward model, electrical reference, conductivity assumptions, and uncertainty	Report sensitivity to source leakage and reference choice
Spatial confounds	Volume-conduction, spatial-leakage, common-input, and distance controls	Withdraw a propagating-wave interpretation if zero-lag or leakage models explain the pattern
Low-amplitude intervals	Preregistered amplitude / SNR exclusion threshold	Do not interpret unstable phase in excluded intervals
Receiver offset	Training-only method for learning ψ_i and its eligible receiver window	No target, test-fold, or future-state information may enter offset selection
Baseline direction	Causal online, causal retrospective, or acausal baseline estimator	A real-time mechanism claim requires a causal estimator; acausal analyses are descriptive
Nonstationarity	State labels, change-point method, drift correction, and reference-update rule	Report failure when one fixed reference cannot generalize across declared states
Feature reduction	Training-only feature selection or dimensionality reduction	Freeze transformation before held-out evaluation and match optimization budgets
Replication	Cross-session, cross-subject, or cross-preparation transfer target	A result confined to one flexible pipeline is exploratory

Appendix D. Statistical and causal-mediation contract

D.1 Confirmatory decision rule

For each primary endpoint, preregistration must state:

1. the estimand and unit of inference;
2. a minimum scientifically meaningful effect;
3. equivalence margins for every variable that must be treated as matched;
4. sample-size or precision justification, including clustering and repeated measures;
5. the null or permutation procedure and exchangeability assumptions;
6. multiplicity control across endpoints, windows, bands, regions, and model families;
7. hierarchical cross-subject or cross-preparation inference;
8. missing-data, artifact, and exclusion handling;
9. the independent-replication threshold; and
10. the exact rule for positive, negative, equivocal, and failed-quality-control outcomes.

Statistical significance without the minimum effect, or absence of significance without an equivalence test, does not satisfy the decision rule.

D.2 Causal mediation

The causal perturbation tests must freeze a directed acyclic graph or an explicitly cyclic time-indexed graph, treatment and mediator time points, total / direct / indirect estimands, and the assumptions needed to identify them. The analysis must examine post-intervention confounding, exposure-mediator interaction, mediator measurement error, treatment-induced selection, and alternative mediator orderings. Negative-control mediators should share nuisance sensitivity without belonging to the proposed mechanism.

The strongest mediation result requires convergence of:

```
targeted perturbation
-> predicted field or PWD change
-> predicted receiver-state interaction
-> downstream neural-state change
-> declared content or self-variable change
```

with either an independent mediator intervention or a mechanism-specific rescue where feasible. Temporal order alone establishes sequence, not mediation.